A1.1 Water

A1 Unity and diversity
Molecules

HL

Understandings:

- A1.1.7—Extraplanetary origin of water on Earth and reasons for its retention
- A1.1.8—Relationship between the search for extraterrestrial life and the presence of water

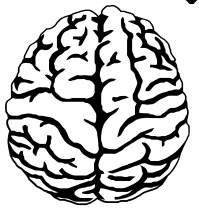
Key words:
Extraplanetary
Extraterrestrial
Goldilocks zone

IB Learner Profile

Inquirers - We nurture our curiosity, developing skills for inquiry and research. We know how to learn independently and with others. We learn with enthusiasm and sustain our love of learning throughout life.



Knowledgeable - We develop and use conceptual understanding, exploring knowledge across a range of disciplines. We engage with issues and ideas that have local and global significance.



Thinkers - We use critical and creative thinking skills to analyse and take responsible action on complex problems. We exercise initiative in making reasoned, ethical decisions



Communicators - We express ourselves confidently and creatively in more than one language and in many ways. We collaborate effectively, listening carefully to the perspectives of other individuals and groups.



Principled - We act with integrity and honesty, with a strong sense of fairness and justice, and with respect for the dignity and rights of people everywhere. We take responsibility for our actions and their consequences.



Open minded - We critically appreciate our own cultures and personal histories, as well as the values and traditions of others. We seek and evaluate a range of points of view, and we are willing to grow from the experience.



Caring - We show empathy, compassion and respect. We have a commitment to service, and we act to make a positive difference in the lives of others and in the world around us.



Risk takers - We approach uncertainty with forethought and determination; we work independently and cooperatively to explore new ideas and innovative strategies. We are resourceful and resilient in the face of challenges and change.



Balanced - We understand the importance of balancing different aspects of our lives—intellectual, physical, and emotional—to achieve well-being for ourselves and others. We recognize our interdependence with other people and with the world in which we live.



Reflective - We thoughtfully consider the world and our own ideas and experience. We work to understand our strengths and weaknesses in order to support our learning and personal development.



A1.1.7—Extraplanetary origin of water on Earth and reasons for its retention


TEDEd
WHERE DID
EARTH'S
WATER
COME FROM?

A1.1.8—Relationship between the search for extraterrestrial life and the presence of water

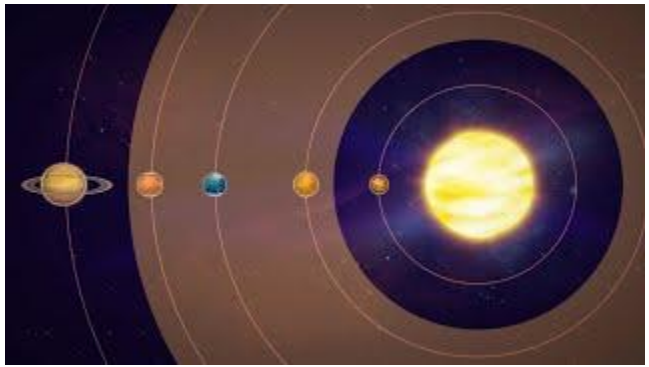
“Goldilocks” zone

A planet that is “just right” for liquid water (and maybe therefore life)

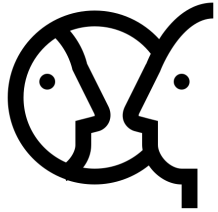
Enough gravity to keep water contained

Not too hot that water would boil away

Not too cold that water would lie frozen



Can you...



Offer a possible explanation for the presence of water on Earth

Explain what is meant by the “goldilocks zone”